

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. Electrical (Electrical Power System) Examination

January 2014

Subject: EE702 Modeling & Simulation of Electrical Machines

Date: 1.1.2014, Wednesday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Draw the generalized mathematical model of a polyphase induction machine. Write down the transformed voltage equations for this model and obtain there from the equivalent circuits for polyphase induction motor. [10]

Q - 2 (a) Prove that the power delivered to any elementary 2 phase machine obtained by transformation of 3 phase to 2 phase, remains same as the power delivered to that of original 3 phase machine. [07]

OR

Q - 2 (a) Draw and explain the block diagram for modeling of Induction machine in details along with various feedback loops, giving the final output as speed and torque of Induction Machine. [07]

Q - 2 (b) Discuss the General Identical transformations of any 3 phase (a, b, c) balanced rotating quantities to 2 phase (q, d). Obtain the transformation matrix, & inverse transformations. [08]

OR

Q - 2 (b) Discuss various inductances of synchronous machines. And derive the equations of the same. [08]

Q - 3 Outline the various steps of applying the generalized theory to the various rotating electrical machines. [10]

OR

Q - 3 (a) Explain the various reference frames and discuss its applications in electrical machines analysis. Give the transformation matrix to transform parameters to stationary reference frames. [04]

(b) Derive the transformation matrix for transforming the variables from one reference frame to another [03]

(c) For D.C. machine, show that the motional inductance $M_d = \frac{\phi Z}{\pi a} * \frac{1}{lf}$ [03]

SECTION – II

Q - 4 Explain the concept of Magnetically coupled circuit applied to operation of Transformers and Electric Machines in details and obtain the inductance matrix for simple magnetically coupled circuits and also draw the equivalent circuit for the same. [10]

Q - 5 (a) Derive the basic energy balance equation for a electromechanical system and explain the same with a example of elementary electromechanical systems. [08]

Q - 5 (b) Obtain the basic set of equations used for modeling of separately excited D.C Generators and deduce the Steady State and Transient analysis of the D.C Generators using the same. [08]

OR

Q - 5 (a) Derive the voltage equations for Kron's primitive machine in matrix form. What observations are made from the impedance matrix of this machine? [08]

Q - 5 (b) Derive the basic set of equations for modeling of D.C. Shunt Machines and explain there from Torque Speed, Torque-Armature Current, and speed-armature current characteristics under steady state conditions if damping constant D is neglected. [08]

Q - 6 Explain the concept, construction and working of a brushless DC machines, along with its advantages and disadvantages of the machines [09]

OR

Q 6 (a) Explain the concept of Speed Voltage and transformer voltages induced in an electromechanical system. [04]

(b) What is Coenergy and Stored Energy, and represent the same Graphically. [05]
